

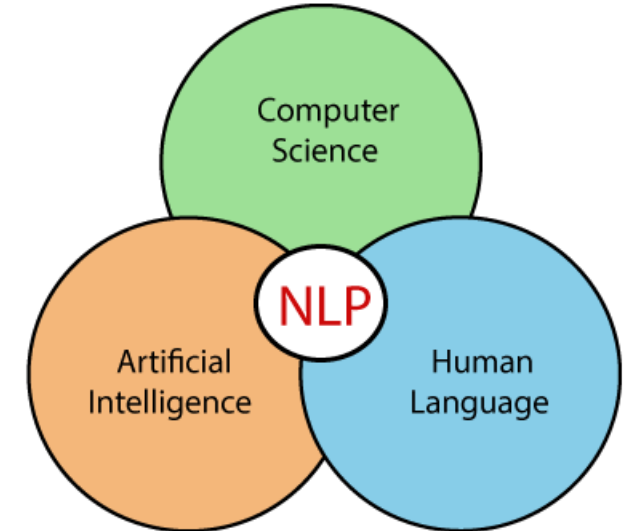


Natural Language Processing

Natural Language Processing

- **What is a Natural Language?**

Any language that has evolved naturally in humans through use and repetition without conscious planning or premeditation.



- **What is a Natural Language Processing?**

A field of computer science, artificial intelligence and computational linguistics concerned with the interactions between computers and human (natural) languages.

The Human Language

[Home](#) / [India](#) / More than 19,500 mother tongues spoken in India: Census

More than 19,500 mother tongues spoken in India: Census

There are 121 languages which are spoken by 10,000 or more people in India, which has a population of 121 crore, the report said.

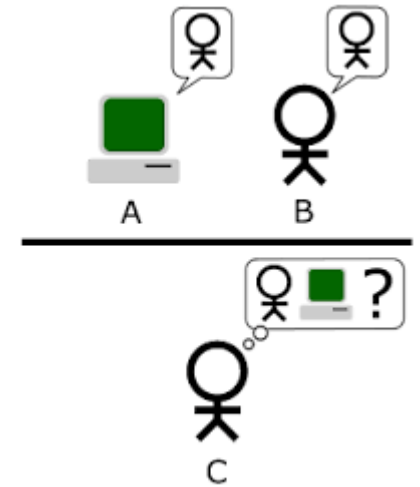
<https://indianexpress.com/article/india/more-than-19500-mother-tongues-spoken-in-india-census-5241056/>



Natural Language Processing

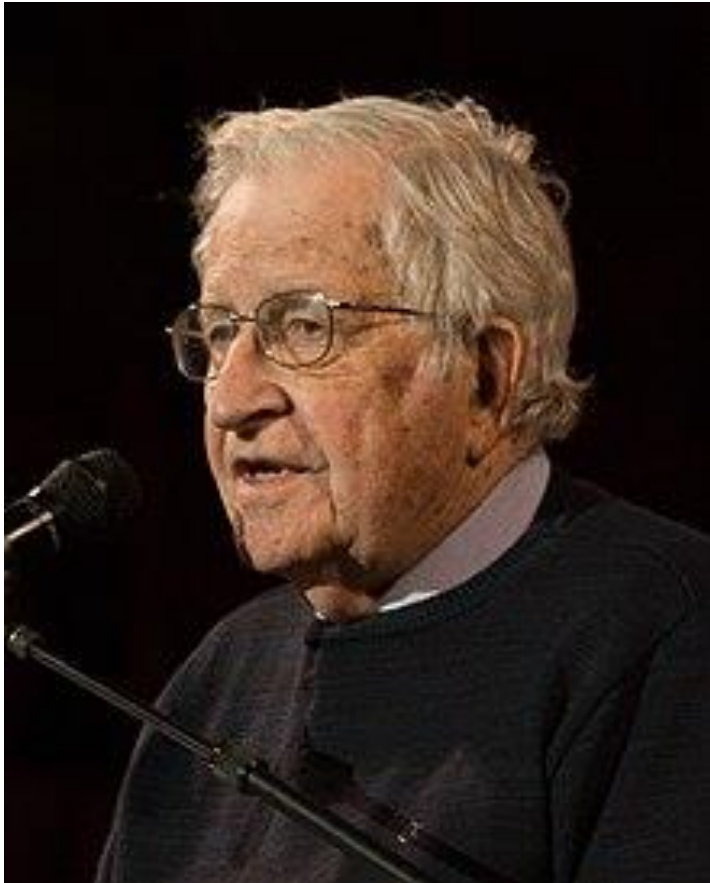


- Setup
 - Two rooms, two humans, and a computer.
 - Room 1: One human C
 - Room 2: One computer (A) and one human (B)
 - A response generated from room 2 (either by A or B)
 - C has to figure out the source of the response
 - If C is successful → "A" failed the turing test
 - Else, → "A" passed the turing test



"[Computing Machinery and Intelligence](#)" which proposed what is now called the [Turing test](#)

Natural Language Processing



The father of modern linguistics

In 1957, **Noam Chomsky's Syntactic Structures** revolutionized Linguistics with '**universal grammar**', a rule based system of syntactic structures

He is a laureate professor of linguistics at the [University of Arizona](#) and an [institute professor](#) emeritus at the [MIT](#)

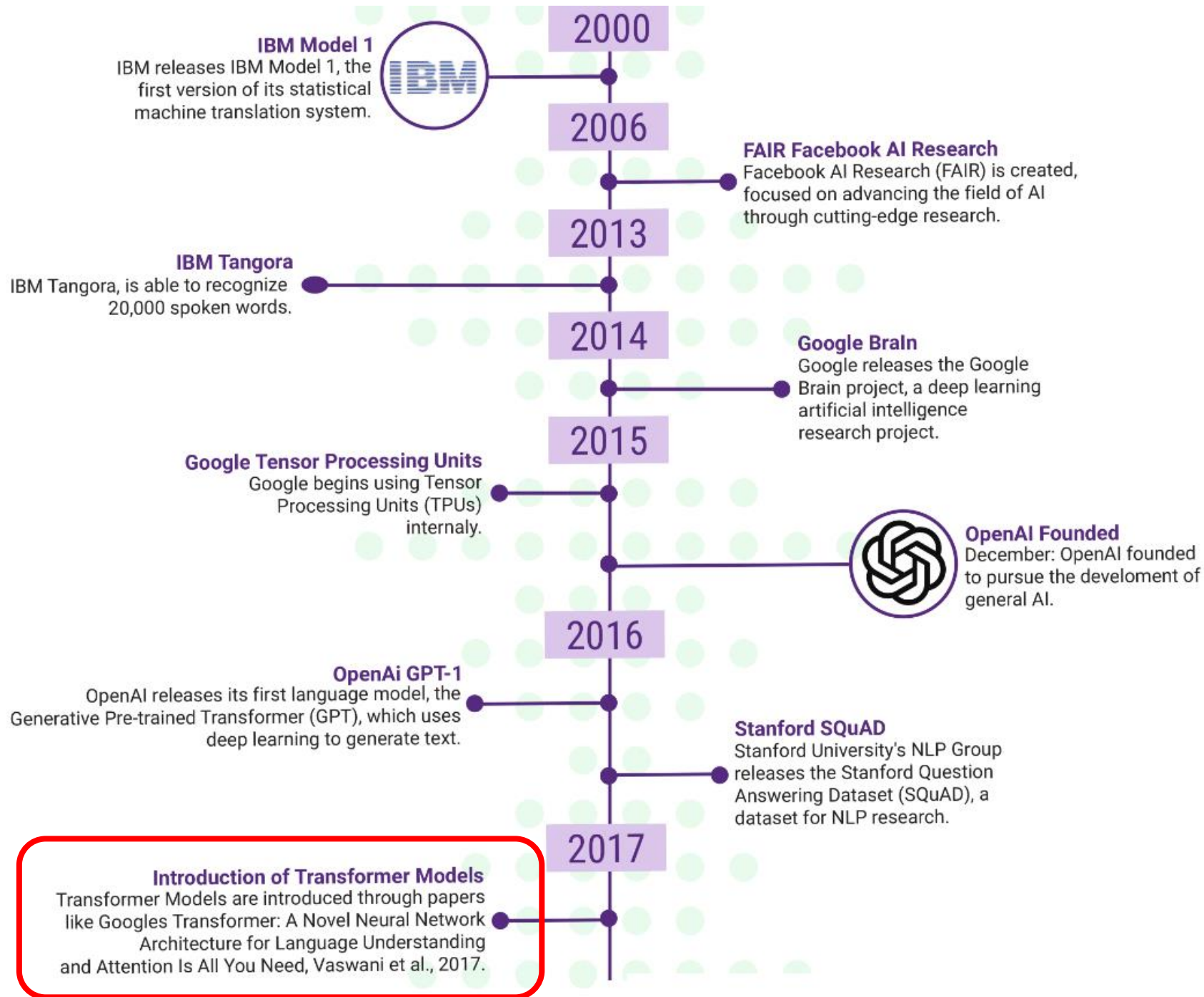
Natural Language Processing

Software	Year	Creator	Description
Georgetown experiment	1954	Georgetown University and IBM	involved fully automatic translation of more than sixty Russian sentences into English.
STUDENT	1964	Daniel Bobrow	could solve high school algebra word problems. ^[6]
ELIZA	1964	Joseph Weizenbaum	a simulation of a Rogerian psychoanalyst, rephrasing her response with a few grammar rules. ^[7]
SHRDLU	1970	Terry Winograd	a natural language system working in restricted "blocks worlds" with restricted vocabularies, worked extremely well
PARRY	1972	Kenneth Colby	A chatterbot
KL-ONE	1974	Sondheimer et al.	a knowledge representation system in the tradition of semantic networks and frames; it is a frame language.
MARGIE	1975	Roger Schank	
TaleSpin (software)	1976	Meehan	
QUALM		Lehnert	
LIFER/LADDER	1978	Hendrix	a natural language interface to a database of information about US Navy ships.
SAM (software)	1978	Cullingford	
PAM (software)	1978	Robert Wilensky	
Politics (software)	1979	Carbonell	
Plot Units (software)	1981	Lehnert	
Jabberwacky	1982	Rollo Carpenter	chatterbot with stated aim to "simulate natural human chat in an interesting, entertaining and humorous manner".
MUMBLE (software)	1982	McDonald	
Racter	1983	William Chamberlain and Thomas Etter	chatterbot that generated English language prose at random.
MOPTRANS ^[8]	1984	Lincoln	
KODIAK (software)	1984	Wilensky	
Absity (software)	1987	Hirst	
Dr. Sbaits	1991	Creative Labs	
Watson (artificial intelligence software)	2006	IBM	A question answering system that won the Jeopardy! contest, defeating the best human players in February 2011.
Siri	2011	Apple	A virtual assistant developed by Apple.
Cortana	2014	Microsoft	A virtual assistant developed by Microsoft.
Amazon Alexa	2014	Amazon	A virtual assistant developed by Amazon.
Google Assistant	2016	Google	A virtual assistant developed by Google.

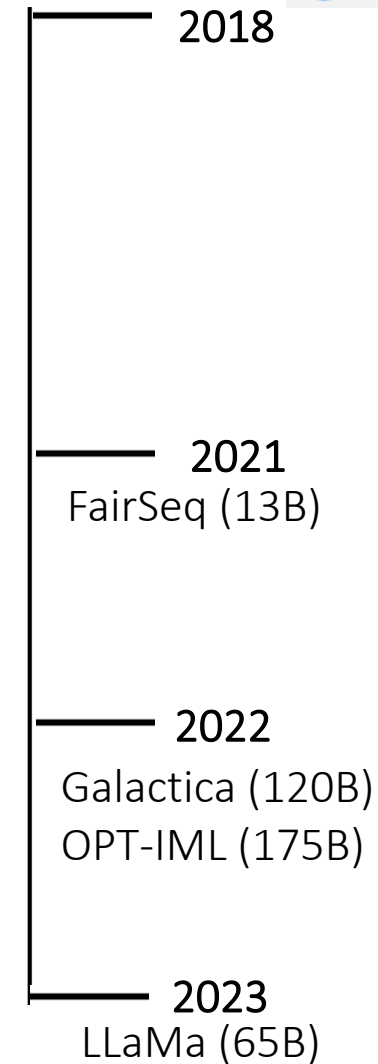
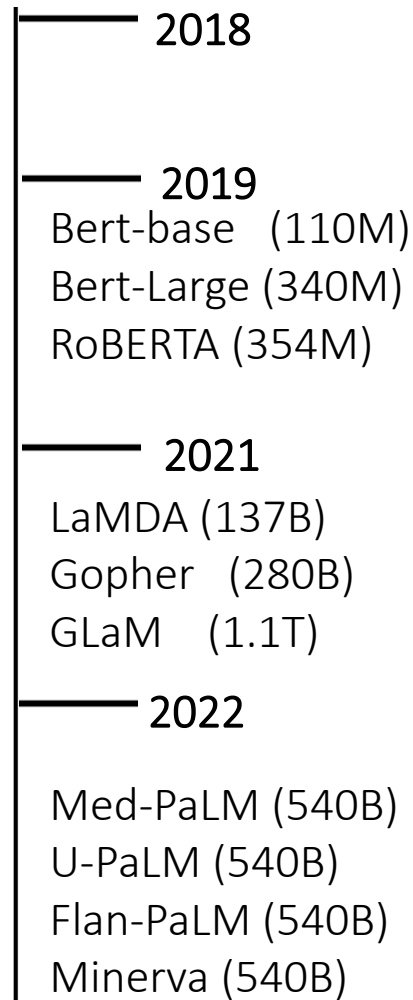
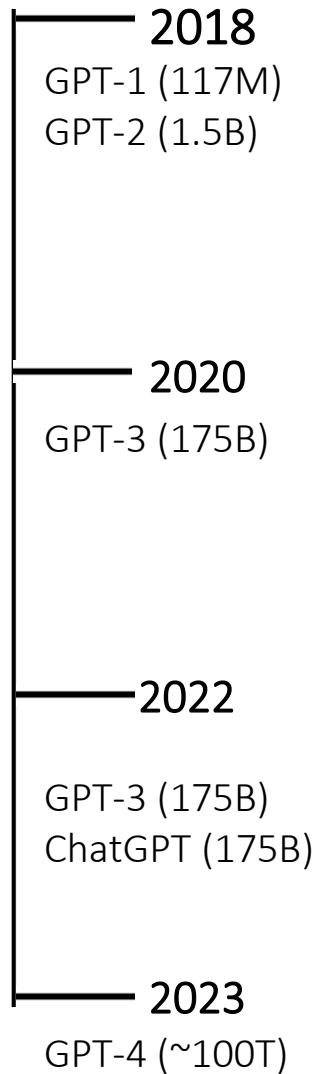
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Timeline History of Large Language Models



Post-Transformer Landscape



Ambiguity

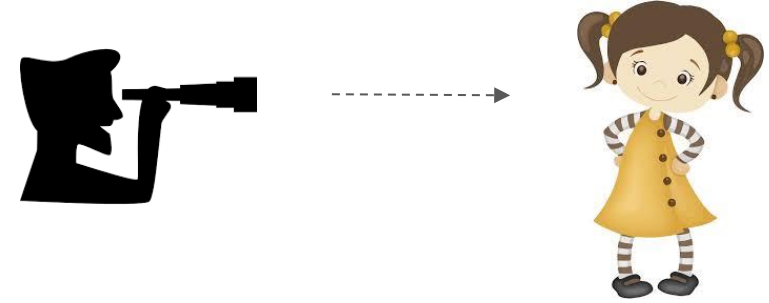
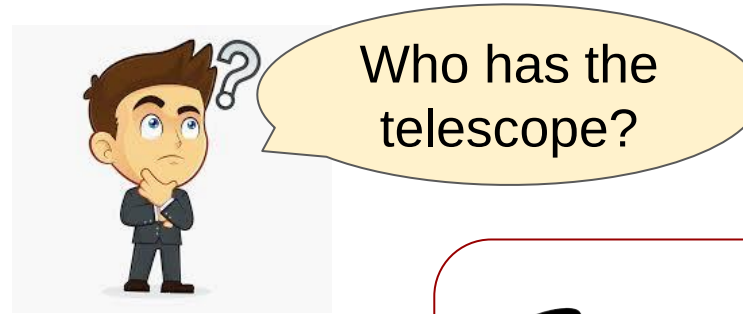
- Is ambiguity present in language only?
 - No, ambiguity is prevalent in every dimension!

Duck or Rabbit?



Ambiguity in language

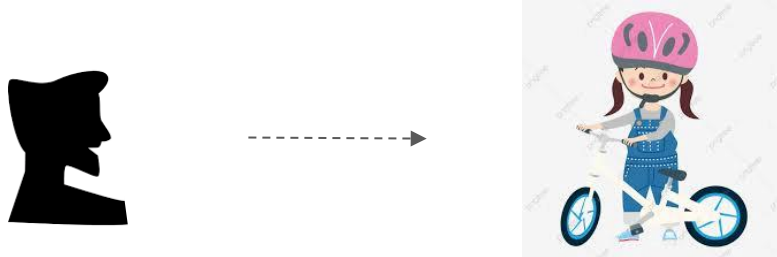
- I saw a girl with a telescope.



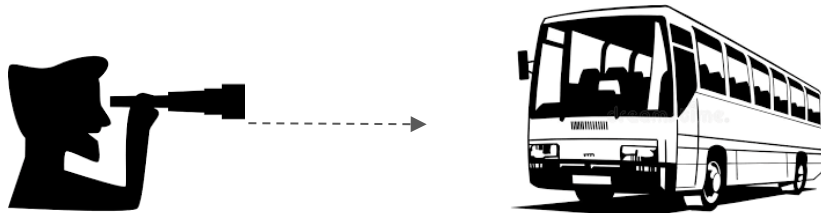
OR



- I saw a girl with a bicycle.



- I saw a bus with a telescope.



No ambiguity!

Ambiguity in language



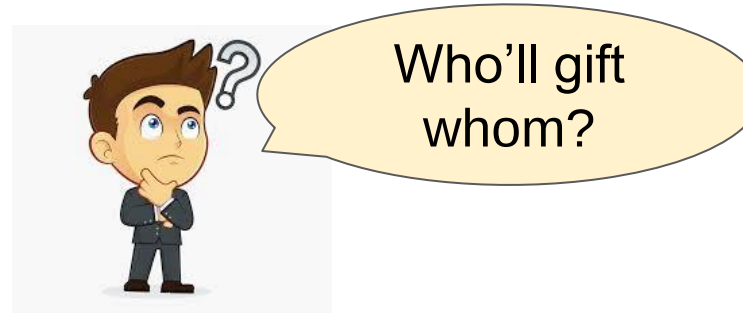
- I saw a girl with a telescope.
- Mary had a little lamb.



OR



Ambiguity in language



- I saw a girl with a telescope.
- Mary had a little lamb.
- Mujhe aapko mithai khilani padegi.

I have to gift you some sweets.

OR

You have to gift me some sweets.

Ambiguity in language

- I saw a girl with a telescope.
- Mary had a little lamb.
- Mujhe aapko mithai khilani padegei.
- **Public demand changes**



OR



- (a) Public demand changes, but does anybody listen to them?
(b) Public demand changes, and we companies have to adapt to such changes.

Ambiguity in language

- I saw a girl with a telescope.
- Mary had a little lamb.
- Mujhe aapko mithai khilani padegi.
- Public demand changes
- I ate rice with spoon.
- I ate rice with curd.
- I ate rice with Rahul.



Similar surface
structures but
different
interpretations!



Ambiguity and Punctuations!

Let's eat Grandma!



Let's eat, Grandma!



A woman without her man is nothing

A woman, without her man, is nothing.

A woman: without her, man is nothing.

Punctuation is powerful.



Components of NLP



**Natural Language
Understanding**



**Natural Language
Generation**

NLP layers

- Understanding the semantics is a non-trivial task.
- Needs to performs a series of incremental tasks to achieve this.
- NLP happens in layers

Pragmatics & Discourse	<i>Study of semantics in context.</i>
Semantics	<i>Meaning of the sentence.</i>
Parsing	<i>Syntactic structure of the sentence.</i>
Chunking	<i>Grouping of meaningful phrases.</i>
Part of speech tagging	<i>Grammatical classes.</i>
Morphology	<i>Study of word structure.</i>



Increasing
Complexity Of
Processing

Word and Token

- Word:
 - Smallest sequence of *phonemes* of a spoken language that can be uttered in isolation
- Word Segmentation/Tokenization:
 - Breaking a string of characters into a sequence of words.
 - Smallest sequence of *graphemes* that are delimited with some predefined characters (space, comma, full-stop, etc.);

Ram, Shyam, and Mohan are playing.

⇒

[Ram] [,] [Shyam] [,] [and] [Mohan] [are] [playing] [.]

21,53,010 COVID cases in India.

⇒

[21] [,] [53] [,] [010] [COVID] [cases] [in] [India] [.]



[21,53,010] [COVID] [cases] [in] [India] [.]



Check this out...https://www.abc.com

⇒

[Check] [this] [out] [.] [.] [.] [https] [:] [/] [/] [www] [.] [abc] [.] [com]



[Check] [this] [out] [...] [https://www.abc.com]



#GreatDayEver

⇒

[#] [Great] [Day] [Ever]

Morphology

- Field of linguistics that studies the internal structure of words
 - How they are formed
 - Their relationship to other words in the same language.
- It defines word formation rule from the root word.
- *Morpheme* is the smallest linguistic unit that has semantic meaning
 - E.g.:
 - “Pre”, “ed”, “ing”, “s”, “es”, etc.
 - Dogs ⇒ dog + s (plural)
 - Going ⇒ go + ing (present participle)
 - Independently ⇒ independent + ly (Adverb)
 - ⇒ in + dependent + ly (Negation)
 - ⇒ in + depend + ent + ly (relying)
 - ⇒ in + de + pend + ent + ly

Pend: (verb) to remain undecided or unsettled.

Morphology

- English, Chinese, etc. are commonly referred as *morphologically-poor* language.
- Indian, Turkish, Hungarian, etc. are termed as *morphologically-rich* language.

English	Hindi	Linguistic property
I will go.	मैं जाऊँगा।	Different morphological forms of word 'will go' in Hindi
We will go.	हम जाएंगे।	
You will go.	तुम जाओगे।	
He will go.	वह जाएगा।	
She will go.	वह जाएगी।	

Parts-of-Speech (POS)

- Grammatical class of the word.

He	ate	an	apple	.
PRP	VBD	DT	NN	.

Tags

PRP: Personal Pronoun

VBD: Verb, Past

DT: Determiner

NN: Noun, Singular, Mass

TO: *to*

IN: Preposition

- PoS disambiguation
 - A word can belong to different grammatical classes.

He	went	to	the	<i>park</i>	in	a	car	.
PRP	VBD	TO	DT	<i>NN</i>	IN	DT	NN	.

They	went	to	<i>park</i>	the	car	in	the	shed	.
PRP	VBD	TO	<i>VB</i>	DT	NN	IN	DT	NN	.

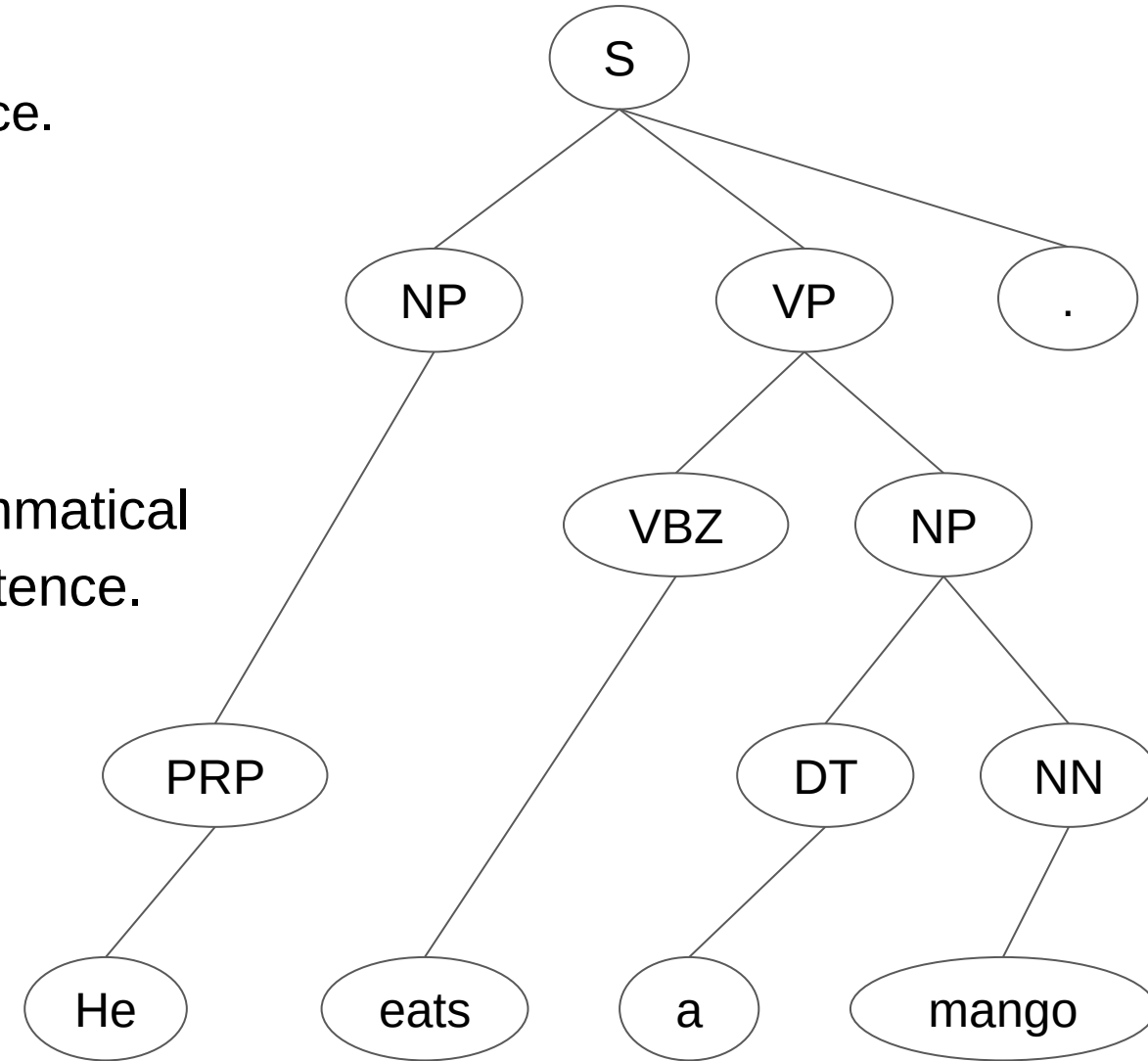
- 45 tags in Penn Treebank tagset
- 146 tags in C7

Chunking

- Identification of non-recursive phrases (noun, verb, etc.)
 - He went to the Indian city Mumbai. ⇒
[NP He] [VP went] [PP to] [NP the Indian city Mumbai]
 - Mumbai green lights women icons on traffic signals earns global praise. ⇒
[NP Mumbai green lights women icons] [PP on] [NP traffic signals] [VP earns] [NP global praise]

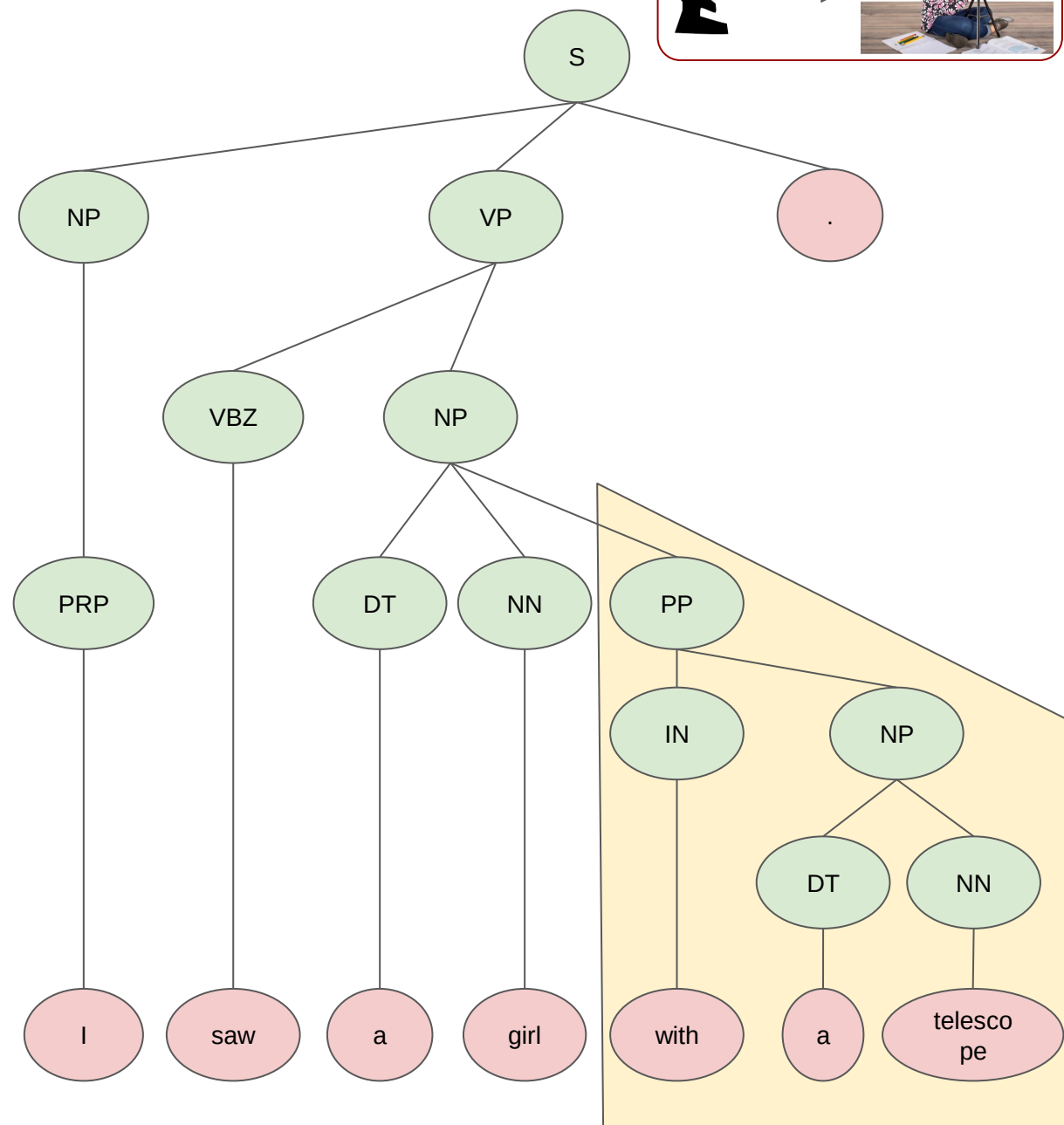
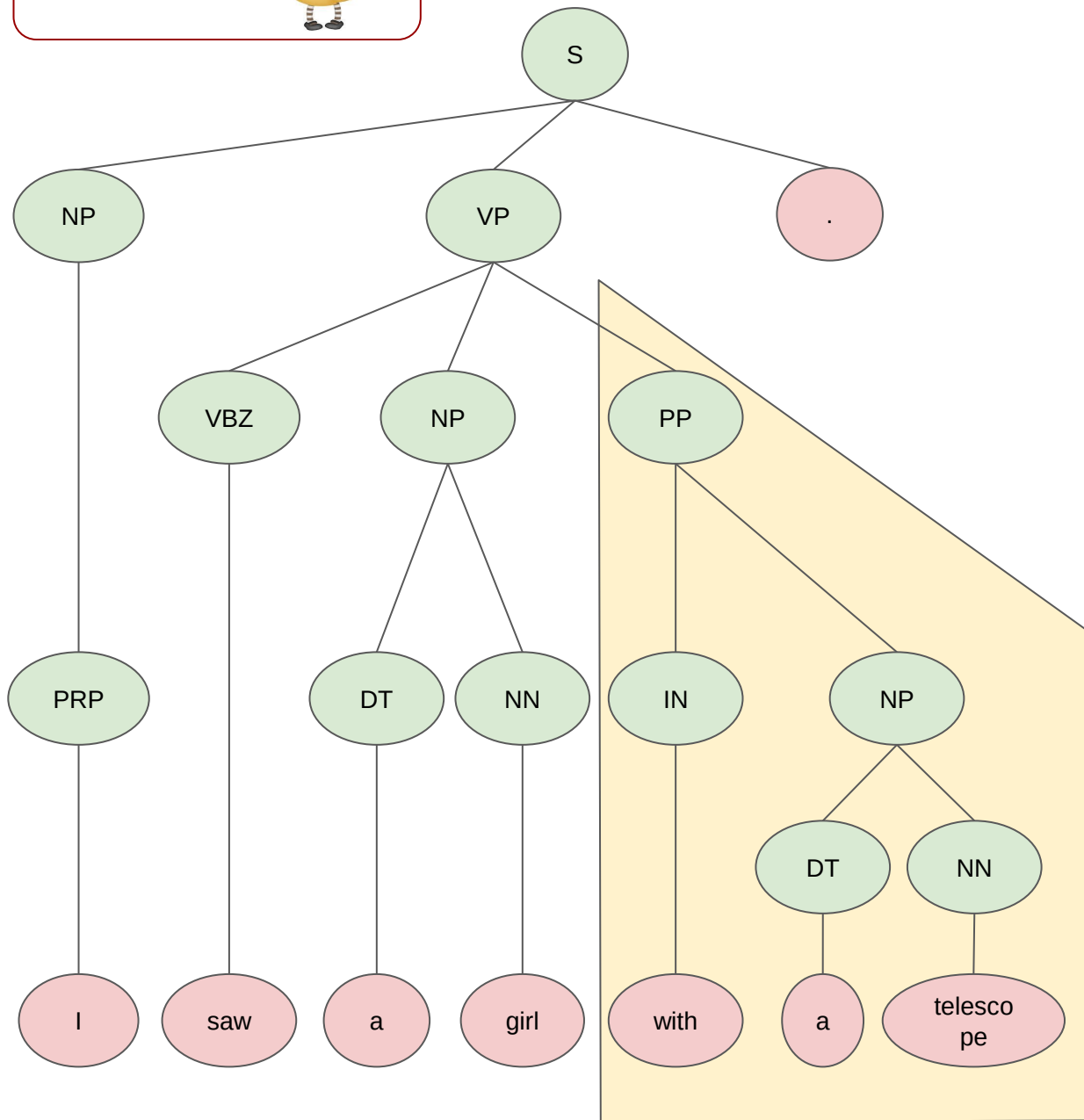
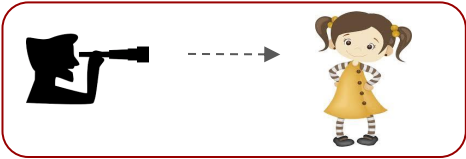
Syntax Processing

- Validate the grammatical structure of the sentence.
- Let, vocabulary = [the, mango, he, eats, ...]
 - He eats a mango. ⇒ 
 - He mango eats a. ⇒ 
- The sequence of words must follow the grammatical structure of the language to form a valid sentence.
 - Construct a parse tree.



Parse Tree

Syntactic Ambiguity



Tasks we want to solve in NLP

Semantic Role Labelling (SRL)

- Identify the semantic role of each argument (noun phrase) w.r.t. the predicate (main verb) of the sentence

John	drove	Mary	from	Delhi	to	Pune	in	his	car
Agent		Patient		source		destination			instrument

Ram	hit	Shyam	with	a	hockey	stick	yesterday
Agent		Patient			instrument		time

Textual Entailment

- Determine whether one natural language sentence entails (implies) another under an ordinary interpretation

(Ram hit Shyam with a hockey stick yesterday. → Shyam got hurt) ⇒ Positive TE

(Ram hit Shyam with a hockey stick yesterday. → Shyam did not get hurt) ⇒ Negative TE

(Ram hit Shyam with a hockey stick yesterday. → Shyam got his first goal) ⇒ non TE

Pragmatics

- Pragmatics considers [Thomas, 1995]:
 - the negotiation of meaning between speaker and listener.
 - the context of the utterance.
 - the intention of the user.
- Context/World knowledge: An employee coming late to the office.
 - Utterance: Do you know what time is it?
 - Literal meaning: Are you aware of the current time? (Response: Yes, it is 12:30 PM)
 - Pragmatic meaning: Why are you coming so late? (Response: Reason for being late.)
- Intention:
 - Utterance: Can you pass the water bottle?
 - Literal meaning: Are you able to pass the water bottle? (Response: Yes, I can.)
 - Pragmatic meaning: Pass me the water bottle. (Response: Handover the water bottle)

Discourse

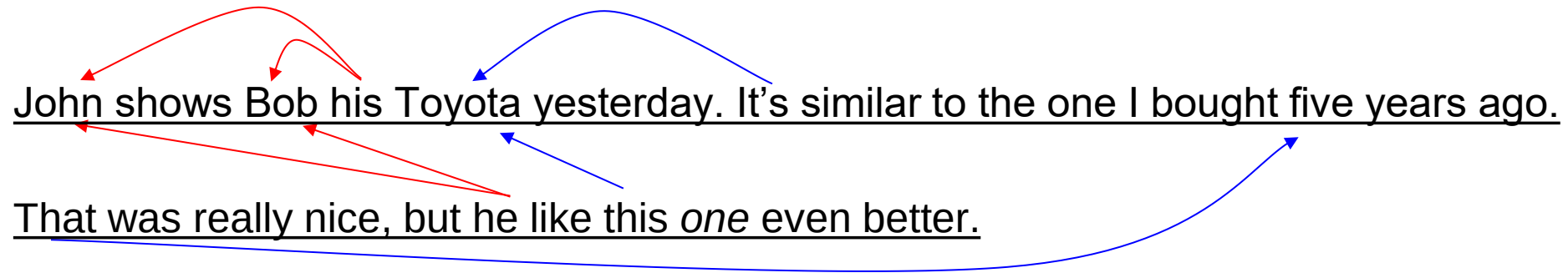
- Processing of sequence of sentences.

Mother said to John: Go to school. It is open today. Are you planning to bunk? Father will be very angry.

- Discourse processing helps answering these questions.
 - What is open?
 - Bunk what?
 - Why the father will be angry?

Coreference Resolution

- Two referring expressions used to refer to the same entity are said to **corefer**.
- Determine which phrases in a document corefer.

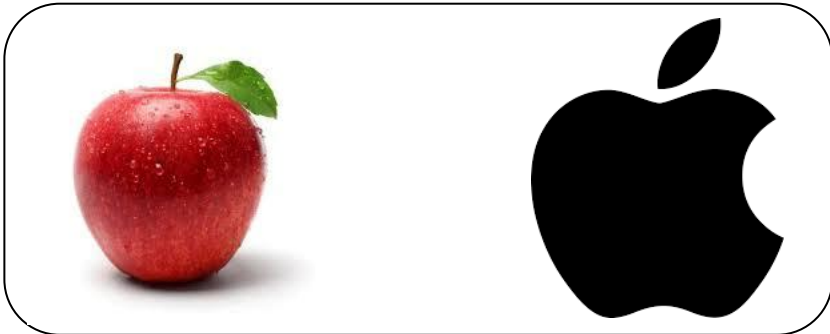


Information Extraction

- Extraction of relevant piece of information
- Named Entity Recognition (NER):
 - Identify names (Proper nouns)
 - [India]_{Location} born [Sundar Pichai]_{Person} is the CEO of [Google]_{Organization} and its parent company [Alphabet]_{Organization}
- Relation extraction:
 - Relation among entities
 - CEO(Sundar Pichai, Google), CEO(Sundar Pichai, Alphabet), Born-at(Sundar Pichai, India), ParentOrg(Alphabet, Google)

Word Sense Disambiguation (WSD)

- What does a word mean?
 - The fisherman went to the *bank*. ⇒ Financial bank or river bank?
 - The fisherman went to the *bank* to withdraw money.
 - The fisherman went to the *bank* to fish.



Sentiment Analysis

- Extract polarity orientation of the subjectivity
 - Really superb pillow. Love to sleep on it.. very comfortable... ⇒ Positive
 - It's a mass Chinese product. Too expensive. Thin and useless ⇒ Negative
 - My neighbours are home and it's good to wake up at 3am in the morning. ⇒ Negative?
 - Campus has deadly snakes. ⇒ Negative
 - Shane Warne is a deadly spinner. ⇒ Positive?
 - The food was cheap. ⇒ Positive?
 - Not to mention the cheap service I got at the restaurant. ⇒ Negative
 - Movie was 4 hrs long. ⇒ Neutral?

Machine Translation

- Given a sentence in the source language L1, convert it to the target language L2, such that the semantic (adequacy and fluency) is preserved.

ENGLISH - DETECTED SOMALI ENG ▼ ↔ HINDI SOMALI ENGLISH ▼

I saw a girl with telescope. × मैंने दूरबीन से एक लड़की को देखा।

English ▼ ↔ Hindi ▼

She is a doctor × वह एक डॉक्टर है
vah ek doktor hai

Hindi ▼ ↔ English ▼

वह एक डॉक्टर है × He is a doctor



Source: Google Translate

Summarization

- Given a document, summarize the semantics (extract relevant information) in shorter length text.
- Document
 - Sen. Barack Obama sealed the Democratic presidential nomination last night after a grueling and history-making campaign against Sen. Hillary Rodham Clinton that will make him the first African American to head a major-party ticket.
- Summary
 - Barack Obama is the Democratic presidential candidate.

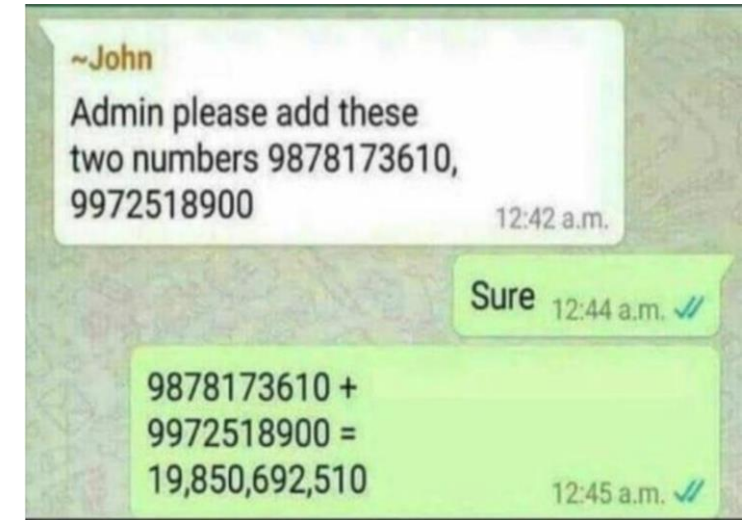
Question Answering

- Answer natural language questions based on information presented in the repository.
- Factoid Questions
 - **Question: Who is the author of the book Wings of Fire?**
 - **Answer: A. P. J, Abdul Kalam**
- List Questions
 - **Question: What are the islands in India?**
 - **Answer: Andaman Island, Nicobar Island, Labyrinth Island, Barren Island**
- Descriptive Questions
 - **Question: What is Greenhouse effect?**
 - **Answer: The analogy used to describe the ability of gases in the atmosphere to absorb heat from the earth's surface.**

Dialog System and Chatbot

- Conversation of two or more parties.

C₁: ...I need to travel in May.
A₁: And, what day in May did you want to travel?
C₂: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₂: And you're flying into what city?
C₃: Seattle.
A₃: And what time would you like to leave Pittsburgh?
C₄: Uh hmm I don't think there's many options for non-stop.
A₄: Right. There's three non-stops today.
C₅: What are they?
A₅: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₆: OK I'll take the 5ish flight on the night before on the 11th.
A₆: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.
C₇: OK.



User:	Do you like Doctor Who?
Cleverbot:	Yes, so funny.
User:	What's your favorite show?
Cleverbot:	I don't have one.
User:	What's your name?
Cleverbot:	Jason I already told you.
User:	No your real name.
Cleverbot:	Stephanie IS my real name.
User:	I thought it was Jason?
Cleverbot:	How did you come to that conclusion.
User:	Because you said so.
Cleverbot:	I have to do my homework.

Why Study NLP?

- To get a job in industry
 - e.g., many current job listings are CL jobs
 - Google Inc.
 - Amazon Inc.
 - Facebook Inc.
 - Flipkart Inc., etc.
- To get a job in academia
 - As a computational linguist
 - computational literacy and an understanding of computational methods will become critical in the next decade.